

Adaptive Kernel Selection for Stein Variational Gradient Descent

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Abstract

Efficient posterior approximation remains a central challenge in Bayesian inference, particularly when uncertainty quantification must be obtained from deterministic particle systems. Stein Variational Gradient Descent (SVGD) addresses this problem by transporting particles toward a target distribution through directions defined in a reproducing kernel Hilbert space (RKHS). Because the kernel determines both the interaction among particles and the kernelized Stein discrepancy (KSD) descent direction, kernel specification is a central modeling choice rather than an incidental tuning parameter. The widely used median heuristic provides a convenient bandwidth rule, but its limited adaptivity can produce severe variance underestimation. This paper introduces Adaptive SVGD (Ad-SVGd), a kernel-selection procedure that updates parameters in a prescribed kernel family by increasing the current KSD during SVGD inference. The method replaces a fixed heuristic bandwidth with an optimization criterion aligned with the discrepancy that governs SVGD descent. We prove, in the discrete-time mean-field setting and under explicit regularity and optimization-accuracy assumptions, convergence of the maximal KSD over the kernel class; a refined result gives KL bounds under a generalized Stein logarithmic Sobolev condition. Experiments on Gaussian mixtures, inverse problems, Bayesian logistic regression, multivariate Gaussian targets, and Gaussian-process inference show that Ad-SVGd consistently improves uncertainty representation relative to median-bandwidth SVGD in the reported settings.

1 Introduction

Stein Variational Gradient Descent (SVGD) [SVGD] is a deterministic particle method for approximate Bayesian inference and an influential alternative to traditional Markov Chain Monte Carlo (MCMC) methods. It transports an ensemble by update directions obtained from the functional gradient of the Kullback-Leibler (KL) divergence to the target distribution, constrained to lie in the unit ball of a reproducing kernel Hilbert space (RKHS). This RKHS constraint makes the kernel a structural component of the algorithm: it specifies the particle interaction, the geometry of the descent direction, and, ultimately, the fidelity of the posterior approximation [Duncan2019OnTG, Nsken2021SteinVG]. A recurring practical failure mode is that finite SVGD particle systems underestimate posterior variance [DBLP:conf/iclr/BaEGSSWZ22], especially as the effective dimension of the approximation problem increases. We demonstrate that this limitation can be mitigated while preserving the original SVGD transport rule. The mechanism is adaptive kernel selection: at successive stages of inference, the kernel parameters are updated by maximizing the kernelized Stein discrepancy (KSD), thereby coupling kernel choice to the discrepancy that drives SVGD descent.

1.1 Related work.

Since its introduction [SVGD], SVGD has become a widely used framework for approximate Bayesian inference in machine learning [DBLP:conf/uai/LiuRLP17, DBLP:conf/iclr/MessaoudMXP0CC2, DBLP:conf/nips/PuGHLHC17, DBLP:journals/tsp/KassabS22]. Its theoretical foundations now include continuous-time mean-field analyses [LLN2019, Duncan2019OnTG, NEURIPS2020'16f8e1], discrete-time convergence results [NEURIPS20203202111c, pmlr-v162-salim22a], finite-particle convergence rates under distinct regularity assumptions [balasubramanian2024improvedfiniteparticleconver, NEURIPS2023'54e5d7af], and gradient-flow interpretations on spaces of probability measures [DBLP:conf/nips/Duncan2019OnTG], in analogy with VariationalFormulationFokker.

Within this literature, kernel selection remains both decisive and incompletely resolved. The kernel controls particle coupling and convergence behavior, as illustrated in Figures 1 and 9; moreover, convergence in KSD implies weak convergence only under conditions that depend on the chosen kernel [pmlr-v70-gorham17a]. The median heuristic [JMLR:v13:gretton12a] is appealing because it is simple and inexpensive, but it lacks a general SVGD-specific justification and can deteriorate as dimensionality increases [DBLP:conf/iclr/BaEGSSWZ22, DBLP:conf/icml/ZhuoLSZCZ18, DBLP:conf/icml/WangZ018]. Several recent methods address high-dimensional degradation by modifying the transport mechanism [stein'newton, NEURIPS2019'5dcd0ddd, gong2021sliced, pmlr-v151-liu22a] or by learning update directions with neural networks [langosco2022neural, ZHAO2023118975]. Our objective is complementary and deliberately focused: we keep the original SVGD particle update fixed and investigate the performance gains that can be obtained by replacing heuristic bandwidth selection with KSD-driven kernel adaptation.

The closest antecedent is the mixture-of-kernels framework of ai'stein'2023, which defines a KSD for a weighted linear combination of base kernels and learns the weights by maximizing the resulting multiple-kernel discrepancy. Ad-SVGd follows the same principle of selecting kernels through KSD, but applies it to continuously parameterized families. This distinction is important for the present study because it makes coordinate-wise and other continuous bandwidth parameterizations directly available without restricting adaptation to a finite basis of preselected kernels.

1.2 Contributions.

This paper asks how much of SVGD's practical variance underestimation can be addressed by optimizing the kernel while leaving the transport rule unchanged. By isolating kernel adaptation from architectural modifications and alternative particle dynamics, the study identifies kernel choice as an independent source of theoretical structure and empirical performance.

Our main contributions are as follows.

- (i) **Adaptive Kernel Selection Method.** We introduce Ad-SVGd, an SVGD variant that updates kernel parameters by maximizing the KSD of the current particle distribution. In contrast to the median heuristic, which supplies a fixed scalar bandwidth rule, the proposed formulation supports continuous multi-parameter adaptation and, in the experiments below, enables coordinate-wise bandwidths.
- (ii) **Theoretical Analysis.** We analyze Ad-SVGd in the discrete-time mean-field setting by extending fixed-kernel KSD descent arguments. The resulting guarantee establishes convergence of the supremal KSD over the parameterized kernel class under an explicit approximation condition for the kernel maximization, and a refined Stein logarithmic Sobolev argument yields

mean-field iteration-complexity bounds.

- (iii) **Empirical Validation.** Across the reported numerical studies, Ad-SVGD improves over median-heuristic SVGD on uncertainty-sensitive posterior-approximation tasks, with the most pronounced gains occurring in settings where the baseline exhibits variance collapse.

2 Mathematical background

We fix a symmetric positive definite kernel $k : \mathbb{R}^d \times \mathbb{R}^d \rightarrow \mathbb{R}$ with associated RKHS $\mathcal{H}_0$. The vector-valued RKHS used by SVGD is the d -fold Cartesian product $\mathcal{H}$, equipped with inner product $\langle f, g \rangle_{\mathcal{H}} = \sum_{i=1}^d \langle f_i, g_i \rangle_{\mathcal{H}_0}$ and canonical feature map $\Phi_k(x) = k(\cdot, x) \in \mathcal{H}_0$. We write $x \cdot y$ for the Euclidean inner product and $\nabla \cdot$ for the divergence operator. The space of probability measures on $(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d))$ is denoted by $\mathcal{P}(\mathbb{R}^d)$, and $\mathcal{P}_p(\mathbb{R}^d)$ denotes the subset with finite p -th moment. For $\mu, \nu \in \mathcal{P}_p(\mathbb{R}^d)$, the Wasserstein p -distance is

$$\mathcal{W}_p(\mu, \nu) = \inf_{\gamma \in \Gamma(\mu, \nu)} \left(\int_{\mathbb{R}^d \times \mathbb{R}^d} \|x - y\|_2^p d\gamma(x, y) \right)^{1/p},$$

where $\Gamma(\mu, \nu)$ is the set of couplings of μ and ν , equivalently the probability measures on $\mathbb{R}^d \times \mathbb{R}^d$ with the prescribed marginals.

2.1 Integral probability metrics and kernelized Stein discrepancy.

Integral probability metrics (IPMs) [IPM] measure discrepancies between probability measures by maximizing differences of expectations over a prescribed class of test functions. Stein’s method [bounderrornorma makes this construction applicable to targets known only up to a normalizing constant by producing target-centered test functions. Specifically, for $\mathcal{S}_\pi f := \nabla \log \pi \cdot f + \nabla \cdot f$ and a suitable kernel, one has $\int_{\mathbb{R}^d} \mathcal{S}_\pi f(x) d\pi(x) = 0$ for all $f \in \mathcal{H}$ [KernelTestGoodness, KernelizedSteinDiscrepancy]. The corresponding *kernelized Stein discrepancy* (KSD) is

$$\text{KSD}(\mu|\pi) := \sup_{f \in B(\mathcal{H})} \left| \int_{\mathbb{R}^d} \mathcal{S}_\pi f d\mu \right|, \quad (1)$$

where $B(\mathcal{H})$ denotes the unit ball in $\mathcal{H}$. The optimizer is $f^* = \frac{\psi}{\|\psi\|_{\mathcal{H}}}$ with $\psi = \int_{\mathbb{R}^d} \mathcal{A}_\pi^k d\mu$, where $\mathcal{A}_\pi^k(x) = \nabla \log \pi(x) \Phi_k(x) + \nabla \Phi_k(x) \in \mathcal{H}$ and Φ_k is the feature map induced by k . Thus the variational problem is equivalently the RKHS norm of the Stein witness function, namely $\text{KSD}(\mu|\pi) = \|\psi\|_{\mathcal{H}}$.

2.2 Stein Variational Gradient Descent.

Given a target distribution π and an initial distribution μ_0 , SVGD constructs transported measures by setting $\mu_{n+1} := T_{\sharp} \mu_n$, where $T_{\sharp} \cdot$ denotes push-forward under $T(x) = x + \gamma \psi^{\mu_n}(x)$. For each $\mu \in \mathcal{P}(\mathbb{R}^d)$, the vector field $\psi^\mu = \int_{\mathbb{R}^d} \mathcal{A}_\pi^k d\mu$ is the RKHS steepest-descent direction for the KL objective. Indeed, the solution of Equation (1) implies that $\frac{\psi^{\mu_n}}{\|\psi^{\mu_n}\|_{\mathcal{H}}}$ minimizes $\frac{d}{d\gamma} \text{KL}(T_{\sharp} \mu_n \| \pi) \Big|_{\gamma=0}$ over the unit ball of $\mathcal{H}$ [SVGD], where $\text{KL}(\cdot \| \cdot)$ denotes KL divergence. Consequently,

$$\frac{d}{d\gamma} \text{KL} \left((\text{Id} + \gamma \psi^{\mu_n})_{\sharp} \mu_n \parallel \pi \right) \Big|_{\gamma=0} = -\text{KSD}^2(\mu_n|\pi). \quad (2)$$

Applying these maps to an initial particle ensemble $\{X_0^i\}_{i=1}^M$ sampled from μ_0 gives the standard finite-particle SVGD discretization

$$X_{n+1}^i = X_n^i + \frac{\gamma}{M} \sum_{j=1}^M k(X_n^i, X_n^j) \nabla \log \pi(X_n^j) + \nabla_{X_n^j} k(X_n^i, X_n^j). \quad (3)$$

3 Adaptive kernel selection for SVGD

Because KL convergence for SVGD is difficult to establish without restrictive assumptions, recent analyses often formulate convergence through the kernelized Stein discrepancy [NEURIPS20203202111c, NEURIPS2023'54e5d7af, pmlr-v162-salim22a]. This perspective makes the role of the kernel explicit: KSD convergence alone need not imply weak convergence, since the induced mean-field measures may fail to be tight, and tightness is necessary for weak convergence in Polish spaces. As Equation (1) shows, the RKHS, and hence the kernel, determines the test-function class through which the target is probed; the selected kernel therefore governs how informative KSD convergence is about convergence of measures [pmlr-v70-gorham17a].

We therefore formulate SVGD over a parameterized kernel family $\{k_\theta \mid \theta \in \Theta\}$. For each $\theta \in \Theta$, let KSD_θ denote the corresponding *kernelized Stein discrepancy*, ψ_θ^μ the associated optimal SVGD direction, and Φ_θ the feature map. This parameterization converts kernel choice into an explicit optimization variable, so that the RKHS geometry can change with the particle distribution rather than remaining tied to a fixed bandwidth rule throughout inference.

The selection criterion is to choose, for the current particle measure, a kernel that maximizes the KSD to the target. This criterion is aligned with the SVGD descent identity: the instantaneous KL decrease is proportional to the squared KSD at the current measure (see Equations (2) and (7)). Maximizing the KSD over the admissible kernel family therefore selects, within that family, the RKHS direction with the largest available first-order decrease. Geometric interpretations of SVGD as a gradient flow support the same principle, since kernels that induce larger KSD values give more favorable convergence behavior when the associated RKHSs are compared [Nskn2021SteinVG, Duncan2019OnTG].

Standard SVGD fixes the kernel and studies convergence of the corresponding KSD. Ad-SVGd instead targets the maximal KSD over a prescribed kernel family. In contrast to the median heuristic of **SVGD**, this formulation applies to continuously parameterized kernel classes for which the KSD can be evaluated and differentiated; at iteration n , the ideal parameter is

$$\theta_n \in \operatorname{argmax}_{\theta \in \Theta} \text{KSD}_\theta(\mu_n | \pi).$$

The implemented algorithm approximates this maximization by taking one or more gradient-ascent steps with step size $s > 0$ on the KSD before applying the SVGD particle update. To reduce computational cost, the user-specified decision variable `paramupdate` permits kernel updates only at selected iterations (see Algorithm 1). The kernel-selection mechanism is modular: the particle transport step can be combined with adaptive step-size schedules, line searches, momentum, or second-order/preconditioned variants, and the KSD ascent step can be replaced by another iterative optimizer. The gradient calculation relies on the explicit KSD representation [**KernelTestGoodness**, **KernelizedSteinDiscrepancy**]

$$\text{KSD}^2(\mu | \pi) = \int_{\mathbb{R}^d \times \mathbb{R}^d} u_\pi^k d(\mu \otimes \mu) \quad (4)$$

with

$$u_{\pi}^k(x, y) = k(x, y) \nabla \log \pi(x) \cdot \nabla \log \pi(y) + \nabla \log \pi(y) \cdot \nabla_x k(x, y) \\ + \nabla \log \pi(x) \cdot \nabla_y k(x, y) + \text{trace}(\nabla_x \nabla_y k(x, y)), \quad (5)$$

from which the kernel-parameter gradient is obtained by differentiating with respect to θ . The additional cost relative to vanilla SVGD is controlled by two implementation features:

- Kernel-parameter ascent reuses the score evaluations $\nabla \log \pi(X_n^i)$ already required by the corresponding SVGD step, so Ad-SVGd does not require additional evaluations of $\nabla \log \pi$.
- The overhead can be moderated by choosing a small number `nstepstheta` of kernel-parameter updates, updating the kernel only after several SVGD steps, or, for large particle ensembles, estimating the KSD from a subsample of particles.

Algorithm 1: Ad-SVGd

Input: Initial particle set $\{X_0^i \in \mathbb{R}^d \mid i = 1, \dots, M\}$, kernel class $\{k_{\theta} \mid \theta \in \Theta\}$, initial kernel parameter $\theta_{-1} \in \Theta$, step sizes $\gamma, s > 0$, number of steps `nsteps`, `nstepstheta` $\in \mathbb{N}$

Output: Final particle set $\{X_{\text{nsteps}}^i \in \mathbb{R}^d \mid i = 1, \dots, M\}$

```

for  $n = 0$  to nsteps  $- 1$  do
    Decide paramupdate  $\in \{\text{True}, \text{False}\}$ ;
    if paramupdate then
         $\theta_n^0 \leftarrow \theta_{n-1}$ ;
        for  $\ell = 0$  to nstepstheta  $- 1$  do
             $\theta_n^{\ell+1} \leftarrow \theta_n^{\ell} + s \nabla_{\theta_n^{\ell}} \text{KSD}_{\theta_n^{\ell}}^2 \left( \frac{1}{M} \sum_{i=1}^M \delta_{X_n^i} \mid \pi \right)$ ;
        end
         $\theta_n \leftarrow \theta_n^{\text{nstepstheta}}$ ;
    else
         $\theta_n \leftarrow \theta_{n-1}$ ;
    end
    for  $i = 1$  to  $M$  do
         $X_{n+1}^i \leftarrow X_n^i + \frac{\gamma}{M} \sum_{j=1}^M k_{\theta_n}(X_n^i, X_n^j) \nabla \log \pi(X_n^j) + \nabla_{X_n^j} k_{\theta_n}(X_n^i, X_n^j)$ ;
    end
end

```

4 Convergence analysis

We consider a target measure with Lebesgue density $\pi(x) \propto \exp(-V)$ for a potential $V : \mathbb{R}^d \rightarrow \mathbb{R}$. The analysis extends recent KSD-based SVGD convergence arguments to the adaptive-kernel setting. Specifically, we adapt the discrete-time mean-field analysis of [pmlr-v162-salim22a], which relies on Talagrand’s inequality rather than the stronger logarithmic Sobolev inequality often used in related work. The assumptions below are the corresponding hypotheses from [pmlr-v162-salim22a], imposed uniformly over the kernel-parameter class.

Assumption 1. *We assume that $V \in C^2$, that $\int_{\mathbb{R}^d} \exp(-V(x)) dx < \infty$, and that the Hessian H of V is uniformly bounded with respect to the operator norm; that is, there exists $L \geq 0$ such that $\|H(x)\|_{\text{op}} \leq L$ for all $x \in \mathbb{R}^d$.*

Assumption 2. We assume that $\pi \in \mathcal{P}_1(\mathbb{R}^d)$ satisfies Talagrand's inequality T1: there exists $\lambda > 0$ such that

$$\mathcal{W}_1(\mu, \pi) \leq \sqrt{2 \text{KL}(\mu \parallel \pi) / \lambda}$$

for all $\mu \in \mathcal{P}_1(\mathbb{R}^d)$.

Assumption 3. We assume that there exists $B > 0$ such that $\|\Phi_\theta(x)\|_{\mathcal{H}_0} \leq B$ for all $x \in \mathbb{R}^d$ and $\theta \in \Theta$, and that $\nabla \Phi_\theta$ is continuous with $\|\nabla \Phi_\theta(x)\|_{\mathcal{H}}^2 = \sum_{i=1}^d \|\partial_i \Phi_\theta(x)\|_{\mathcal{H}_0}^2 \leq B^2$ for all $x \in \mathbb{R}^d$ and $\theta \in \Theta$.

4.1 Convergence analysis of SVGD under Talagrand's inequality

We first recall the fixed-kernel descent mechanism that underlies the adaptive analysis. Throughout this subsection, set

$$\mathcal{F}(\mu) := \text{KL}(\mu \parallel \pi).$$

The proof uses the following fundamental inequality. Fix a kernel parameter $\theta \in \Theta$ satisfying Assumption 3 and, for arbitrary $g \in \mathcal{H}$, define the pushforward measure

$$\tilde{\mu} = (I + \gamma g) \# \mu$$

. Under Assumptions 1 and 3, if $\gamma, B > 0$, $\alpha > 1$, and $g \in \mathcal{H}_\theta$ satisfy $\gamma \|g\|_{\mathcal{H}_\theta} \leq \frac{\alpha-1}{\alpha B}$, Proposition 3.1 of [pmlr-v162-salim22a] gives

$$\mathcal{F}(\tilde{\mu}) \leq \mathcal{F}(\mu) + \gamma \langle \psi_\theta^\mu, g \rangle_{\mathcal{H}_\theta} + \frac{\gamma^2 K}{2} \|g\|_{\mathcal{H}_\theta} \quad (6)$$

with $K = (\alpha^2 + L)B$. Applying this inequality to the SVGD direction yields the fixed-kernel descent condition for the iterates (3) [pmlr-v162-salim22a],

$$\mathcal{F}(\mu_{n+1}) \leq \mathcal{F}(\mu_n) - \gamma \left(1 - \frac{\gamma B(\alpha^2 + L)}{2}\right) \text{KSD}_\theta^2(\mu_n | \pi), \quad (7)$$

provided that

$$\gamma \leq (\alpha - 1) \left(\alpha B (1 + \|\nabla V(0)\|) + L \int_{\mathbb{R}^d} \|x\| d\pi(x) + L \sqrt{\frac{2\mathcal{F}(\mu_0)}{\lambda}} \right)^{-1}. \quad (8)$$

The central verification step is the bound $\gamma \|\psi_\theta^{\mu_n}\|_{\mathcal{H}_\theta} \leq \frac{\alpha-1}{\alpha B}$, obtained from Talagrand's inequality in Assumption 2; this bound permits the use of (6) with $g = \psi_\theta^{\mu_n}$. The descent condition (7) then implies

$$\lim_{n \rightarrow \infty} \text{KSD}_\theta^2(\mu_n | \pi) = 0,$$

because $\sum_{n=0}^{\infty} \text{KSD}_\theta^2(\mu_n | \pi) \leq c_\gamma^{-1} \mathcal{F}(\mu_0)$ for $c_\gamma = \gamma(1 - \frac{\gamma B(\alpha^2 + L)}{2}) > 0$.

For an adaptive parameter sequence (θ_n) , the same descent calculation applies at each iteration with the current kernel parameter and gives

$$\mathcal{F}(\mu_{n+1}) \leq \mathcal{F}(\mu_n) - \gamma \left(1 - \frac{\gamma B(\alpha^2 + L)}{2}\right) \text{KSD}_{\theta_n}^2(\mu_n | \pi).$$

4.2 Convergence analysis of Ad-SVGD.

We first analyze the idealized adaptive iteration in which KSD maximization over the kernel class is solved exactly:

$$\begin{aligned}\mu_{n+1} &= (I + \gamma \psi_{\theta_n}^{\mu_n})_{\#} \mu_n, \\ \theta_n &\in \operatorname{argmax}_{\theta \in \Theta} \operatorname{KSD}_{\theta}^2(\mu_n | \pi),\end{aligned}\tag{9}$$

Equivalently,

$$\|\psi_{\theta_n}^{\mu_n}\|_{\mathcal{H}_{\theta_n}} = \max_{\theta \in \Theta} \operatorname{KSD}_{\theta}(\mu_n | \pi).$$

This exact formulation is available when KSD maximization over the kernel parameter has a closed-form solution, as in the multiple-kernel SVGD framework of [ai'stein'2023], where the kernel is a convex combination of base kernels

$$k_{\theta}(x, y) = \sum_{i=1}^N \theta_i k_i(x, y) \quad \text{with} \quad \sum_{i=1}^N \theta_i = 1.$$

For general continuously parameterized families, the update θ_n in (9) must be approximated. In this abstract formulation, we assume that $\operatorname{argmax}_{\theta \in \Theta} \operatorname{KSD}_{\theta}^2(\mu_n | \pi) \neq \emptyset$ for all $n \in \mathbb{N}$.

The corresponding descent statement is a direct consequence of Theorem 3.2 of [pmlr-v162-salim22a]:

Lemma 1. *Suppose that Assumptions 1-3 are satisfied. For any $\alpha > 1$ with*

$$\gamma \leq (\alpha - 1) \left(\alpha B(1 + \|\nabla V(0)\| + L \int_{\mathbb{R}^d} \|x\| d\pi(x) + L \sqrt{\frac{2\mathcal{F}(\mu_0)}{\lambda}}) \right)^{-1},$$

there exists $c_{\gamma} > 0$ such that for all $n \in \mathbb{N}$

$$\mathcal{F}(\mu_{n+1}) \leq \mathcal{F}(\mu_n) - c_{\gamma} \max_{\theta \in \Theta} \operatorname{KSD}_{\theta}^2(\mu_n | \pi),$$

where $(\mu_n)_{n \in \mathbb{N}}$ is generated by (9).

Corollary 2. *Under the same assumptions as in Lemma 1 it holds that $\lim_{n \rightarrow \infty} \max_{\theta \in \Theta} \operatorname{KSD}_{\theta}^2(\mu_n | \pi) = 0$.*

The implementable algorithm only approximately solves the KSD maximization problem. Suppose that $\max_{\theta \in \Theta} \operatorname{KSD}_{\theta}(\mu | \pi) < \infty$ for every $\mu \in \mathcal{P}(\mathbb{R}^d)$ and consider the alternating scheme

$$\begin{aligned}\theta_n &= \Psi_n(\theta_{n-1}, \mu_n), \\ \mu_{n+1} &= (\operatorname{Id} + \gamma \psi_{\theta_n}^{\mu_n})_{\#} \mu_n\end{aligned}\tag{10}$$

where $\Psi_n : \Theta \times \mathcal{P}(\mathbb{R}^d) \rightarrow \Theta$ denotes an iterative kernel-parameter update intended to increase $\operatorname{KSD}_{\theta}(\mu_n | \pi)$. In Algorithm 1, Ψ_n is implemented by gradient ascent on the KSD. The theorem below does not prove that this particular ascent scheme satisfies the required accuracy condition; instead, it isolates the optimization property under which the adaptive descent argument is valid and thereby also covers other suitable kernel-optimization routines.

Assumption 4. We assume that there exists a sequence $(\varepsilon_n)_{n \in \mathbb{N}}$ such that $\lim_{n \rightarrow \infty} \varepsilon_n = 0$ and

$$\max_{\theta \in \Theta} \text{KSD}_{\theta}^2(\mu_n | \pi) - \text{KSD}_{\theta_n}^2(\mu_n | \pi) \leq \varepsilon_n \quad \text{for all } n \in \mathbb{N}.$$

This approximation condition is sufficient to recover convergence of the maximal KSD over the prescribed kernel class.

Theorem 3. Suppose that Assumptions 1-3 are satisfied. Under Assumption 4, for any $\alpha > 1$ with

$$\gamma \leq (\alpha - 1) \left(\alpha B(1 + \|\nabla V(0)\| + L \int_{\mathbb{R}^d} \|x\| d\pi(x) + L \sqrt{\frac{2\mathcal{F}(\mu_0)}{\lambda}}) \right)^{-1},$$

it holds that $\lim_{n \rightarrow \infty} \max_{\theta \in \Theta} \text{KSD}_{\theta}(\mu_n | \pi) = 0$, where $(\mu_n)_{n \in \mathbb{N}}$ is generated by (10).

Proof. Theorem 3.2 in [pmlr-v162-salim22a] gives

$$\mathcal{F}(\mu_{n+1}) \leq \mathcal{F}(\mu_n) - c_{\gamma} \text{KSD}_{\theta_n}^2(\mu_n | \pi)$$

Because $(\mathcal{F}(\mu_n))_{n \in \mathbb{N}}$ is monotone decreasing and bounded below by zero, it converges and is therefore Cauchy. Hence,

$$c_{\gamma} \text{KSD}_{\theta_n}^2(\mu_n | \pi) \leq \mathcal{F}(\mu_n) - \mathcal{F}(\mu_{n+1}) \rightarrow 0$$

as $n \rightarrow \infty$. Under Assumption 4 we obtain

$$\max_{\theta \in \Theta} \text{KSD}_{\theta}^2(\mu_n | \pi) \leq \text{KSD}_{\theta_n}^2(\mu_n | \pi) + \varepsilon_n \rightarrow 0$$

as $n \rightarrow \infty$. □

A Stein logarithmic Sobolev inequality strengthens this KSD convergence result to a KL convergence statement. Under explicit regimes for the approximation error ε_n , Appendix B derives iteration-complexity bounds for the mean-field system (10) and quantifies the residual bias when each KSD maximization is solved only to fixed $\bar{\varepsilon}$ accuracy. The generalized condition used there requires only one kernel in the class $\{k_{\theta}\}$ to satisfy the Stein logarithmic Sobolev inequality, which preserves flexibility in the choice of adaptive kernel family while making the scope of the guarantee explicit.

5 Numerical Experiments

We evaluate Ad-SVGD on posterior-approximation tasks chosen to isolate the effect of kernel adaptation: a one-dimensional Gaussian mixture model, a linear inverse problem governed by an ordinary differential equation, and Bayesian logistic regression. Additional experiments in Appendix A examine scaling for Gaussian targets and Gaussian-process inference. Throughout, the principal comparison is between Ad-SVGD and SVGD with the median heuristic, denoted Med-SVGD, using the same underlying particle-update structure.

5.1 Kernel parameterization

SVGD is commonly implemented with kernels of the form

$$k_h(x, y) = \exp\left(-\frac{\|x - y\|_p^p}{h}\right),$$

where $\|\cdot\|_p$ is the p -norm on $\mathbb{R}^d$ [SVGD, DBLP:conf/iclr/BaEGSSWZ22, Duncan2019OnTG]. The scalar parameter h is the kernel *bandwidth*. The median heuristic sets $h = \frac{\text{med}^p}{\log(M-1)}$, where med is the current median inter-particle distance, with the intended effect of balancing the attraction and repulsion terms in the SVGD update (3) [SVGD].

To use the multi-parameter structure of Ad-SVG, the experiments employ product kernels

$$k_h(x, y) = \prod_{i=1}^d \exp\left(-\frac{|x_i - y_i|^p}{h_i}\right)$$

with $h = (h_1, \dots, h_d)$, so that each coordinate has its own bandwidth. The derivatives required by Algorithm 1 follow directly from Equations (4) and (5). We also considered a coordinate-wise median heuristic for these product kernels, but this direct extension did not yield reliable approximations and exhibited variance collapse.

The product kernels used here provide a controlled test of whether continuous bandwidth adaptation improves the original SVGD dynamics. More scalable parameterizations, such as spectral or low-rank kernels, could adapt dominant directions of variability without the full cost of coordinate-wise products, while Matérn or mixed-product constructions would provide additional control over smoothness and anisotropy.

5.2 Toy example

We first revisit the one-dimensional Gaussian-mixture example from SVGD, with target $\pi = \frac{1}{3}\mathcal{N}(-2, 1) + \frac{2}{3}\mathcal{N}(2, 1)$. We initialize $M \in \{50, 200, 500\}$ particles from $\mathcal{N}(0, 1)$ and run SVGD for 10^4 iterations with step size 1, using the kernels above with $p = 1$ and fixed choices of bandwidth h . Sample quality is measured by the Wasserstein 1-distance $\mathcal{W}_1$, computed from the explicit one-dimensional formula $\mathcal{W}_1(\mu, \nu) = \int_{\mathbb{R}} |F_\mu(x) - F_\nu(x)| dx$ (see StatisticalAspectsWasserstein) using an exact sample of size 10^5 . As Figure 1 shows, fixed-bandwidth SVGD performs well only over a restricted range of h . This controlled example makes the empirical motivation transparent: even in one dimension, bandwidth is performance-critical, and a robust implementation benefits from a selection rule tied to the objective rather than to a static heuristic.

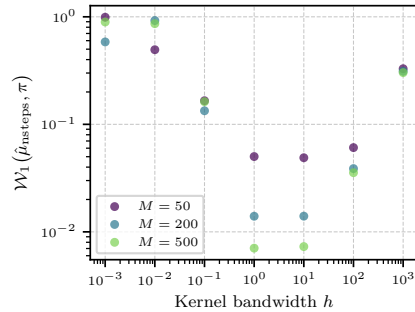


Figure 1: Final Wasserstein 1-distances in the one-dimensional Gaussian-mixture example for fixed-bandwidth SVGD across choices of h .

5.3 Linear inverse problem based on ODE

The next experiment, adapted from [Weissmann2022Optimization], evaluates posterior uncertainty recovery in a linear inverse problem. We consider the one-dimensional differential equation

$$\begin{cases} -f''(s) + f(s) = u(s) & \text{for } s \in (0, 1) \\ f(s) = 0 & \text{for } s \in \{0, 1\} \end{cases} \quad (11)$$

and the associated inverse problem of recovering the forcing term $u(\cdot) \in L^2([0, 1])$ from noisy point evaluations of the solution $f \in H^2([0, 1]) \cap H_0^1([0, 1])$. The observation model is

$$y = \Phi(u) + \varepsilon \in \mathbb{R}^{N_{\text{obs}}},$$

where $\varepsilon \in \mathbb{R}^{N_{\text{obs}}}$ is observational noise. The forward map $\Phi : L^2([0, 1]) \rightarrow \mathbb{R}^{N_{\text{obs}}}$ is $\mathcal{O} \circ H^{-1}$, with $H(f) = -f'' + f$ and observation operator $\mathcal{O}(f) = (f(s_1), \dots, f(s_{N_{\text{obs}}}))^\top \in \mathbb{R}^{N_{\text{obs}}}$ evaluated at the equidistant points $s_k = \frac{k}{N_{\text{obs}}}$, $k = 1, \dots, N_{\text{obs}}$.

For the Bayesian formulation, u is assigned a Gaussian-process (GP) prior represented by the truncated Karhunen-Loève (KL) expansion

$$u(\cdot, x) = Ax = \sum_{k=1}^{N_x} x_k \psi_k,$$

where $\psi_k(s) = \sqrt{2} \sin(\pi k s)$ and $x_k \sim \mathcal{N}(0, \lambda_k)$ independently with $\lambda_k = 50k^{-2}$.

The inference problem is therefore finite-dimensional: estimate the KL coefficient vector $x = (x_1, \dots, x_{N_x})^\top \in \mathbb{R}^{N_x}$ under the Gaussian prior $\mathcal{N}(0, \Gamma_0)$ with $\Gamma_0 = \text{diag}(\lambda_1, \dots, \lambda_{N_x})$. With $\varepsilon \sim \mathcal{N}(0, \Gamma)$, the posterior density is

$$\pi(x) \propto \exp \left(-\frac{1}{2} \|\Gamma^{-1/2}(y - \Phi Ax)\|^2 - \frac{1}{2} \|\Gamma_0^{-1/2} x\|^2 \right), \quad x \in \mathbb{R}^{N_x}.$$

In the implementation, H is replaced by a numerical discretization of Equation (11) on a grid with mesh size 2^{-8} , and the system is fully observed, so $N_{\text{obs}} = 2^8$. We use $N_x = 16$ KL terms and noise covariance $\Gamma = 10^{-3} \text{Id}_{N_{\text{obs}}}$. Reference observations $\bar{y} \in \mathbb{R}^{N_{\text{obs}}}$ are generated by drawing $\bar{x} \sim \mathcal{N}(0, \Gamma_0)$ and setting $\bar{y} = \Phi A \bar{x}$.

We compare Med-SVGD and Ad-SVGD for particle ensemble sizes $M \in \{50, 100, 200\}$ using the kernels in Section 5.1 with $p = 1$. Both methods are run for $4 \cdot 10^5$ iterations with particle step size 10^{-3} and the AdaGrad-style adaptive step-size control suggested by **SVGD**. For Ad-SVGD, the bandwidth step size is 10^{-5} , and the bandwidth is updated once every 100 particle updates. Under this update schedule, the two methods have no significant runtime difference in this experiment. Figure 2 shows the GP reconstruction for one representative seed. Both methods recover the posterior mean, but Ad-SVGD more accurately represents posterior uncertainty. We quantify approximation quality by the Wasserstein 2-distance $\mathcal{W}_2(\mathcal{N}(\hat{\mu}, \hat{\Sigma}), \pi)$ between the target posterior π and the Gaussian approximation $\mathcal{N}(\hat{\mu}, \hat{\Sigma})$, where $\hat{\mu}$ and $\hat{\Sigma}$ are the particle sample mean and

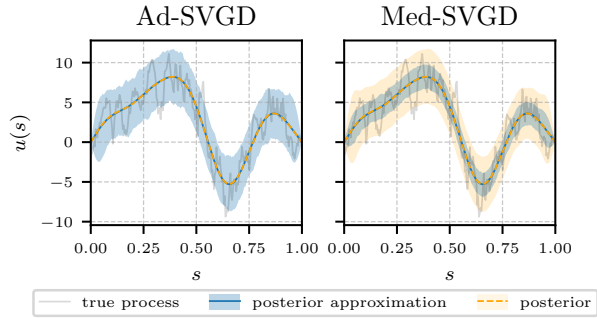


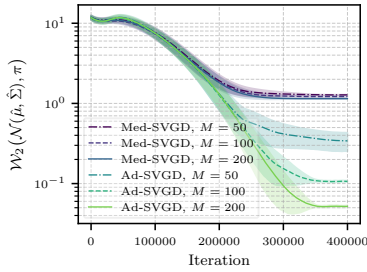
Figure 2: Gaussian-process reconstruction for the ODE-based inverse problem, showing the posterior mean and 90% confidence interval.

With $\varepsilon \sim \mathcal{N}(0, \Gamma)$, the posterior density

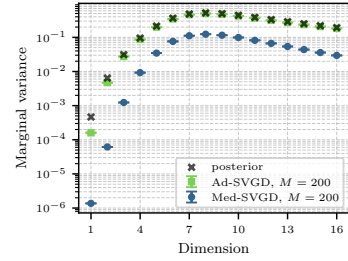
covariance [StatisticalAspectsWasserstein]. Because π is multivariate normal, this distance has the explicit form

$$\mathcal{W}_2(\mathcal{N}(\hat{\mu}, \hat{\Sigma}), \pi)^2 = \|\hat{\mu} - \mu_\pi\|^2 + \text{trace}\left(\hat{\Sigma} + \Sigma_\pi - 2(\hat{\Sigma}^{1/2}\Sigma_\pi\hat{\Sigma}^{1/2})^{1/2}\right),$$

where μ_π and Σ_π are the posterior mean and covariance. We also compare the marginal variances of the final particle distribution with the exact posterior variances. Figure 3 reports the results over 56 random seeds; the posterior covariance is independent of $\bar{y}$. The aggregate results show that Ad-SVGD gives a more accurate Gaussian posterior approximation than Med-SVGD, which systematically underestimates uncertainty. Unlike Med-SVGD, Ad-SVGD continues to improve as the particle number increases beyond 50. Figure 4 further shows that the learned component-wise bandwidths stabilize earlier than the approximation error and differ substantially across coordinates, with final bandwidths negatively correlated with the corresponding marginal variances.

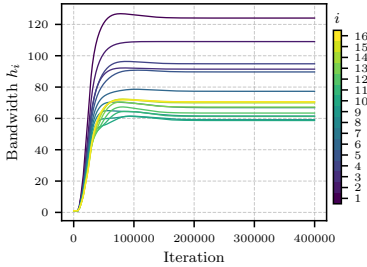


(a) Convergence of approximation error

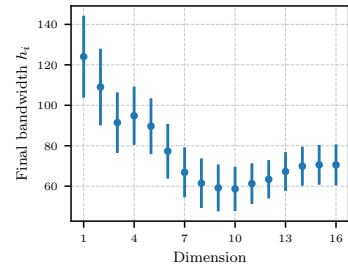


(b) Marginal variances

Figure 3: ODE-based inverse problem: approximation error and marginal variances for Med-SVGD and Ad-SVGD, reported as means with 95% confidence intervals over 56 random seeds.



(a) Evolution of bandwidth per dimension (mean)



(b) Final bandwidths (mean and 95% conf. interval)

Figure 4: ODE-based inverse problem: evolution and final values of the Ad-SVGD component-wise bandwidth parameters, aggregated over 56 random seeds.

5.4 Bayesian logistic regression

Following **SVGD** and **pmlr-v151-liu22a**, we next consider Bayesian logistic regression (BLR) on the Covertypes data set [covertypes]. The regression weights w have prior $w \mid \alpha \sim \mathcal{N}(0, \alpha^{-1})$ with $\alpha \sim \text{Gamma}(1, 0.01)$, and the inferential target is the posterior of $x = [w, \alpha]$. As in

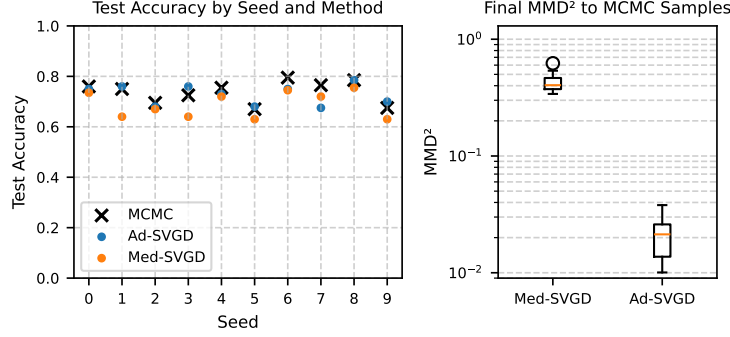


Figure 5: Bayesian logistic regression: SVGD particle quality measured by prediction accuracy (left) and MMD^2 to NUTS reference samples (right).

pmlr-v151-liu22a, we use the No-U-Turn Sampler (NUTS) of **nuts** to generate reference samples, enabling evaluation of both predictive performance and posterior approximation quality for Med-SVGd and Ad-SVGd. Following **pmlr-v151-liu22a**, we evaluate both methods on random subsets of 1000 observations and aggregate over 10 draws. Classification performance is measured by test-set prediction accuracy using the particle mean. Figure 5 shows that Ad-SVGd outperforms Med-SVGd for almost all seeds, attaining higher test accuracy and producing particle distributions closer to the MCMC reference.

Posterior approximation quality is also measured in Figure 5 by the squared maximum mean discrepancy (MMD^2) [**MMDref**] to the MCMC samples. The two methods differ most clearly in their representation of posterior uncertainty. Figure 7 compares the covariance matrices of the final particles with the MCMC reference, averaged over 10 seeds and displayed in the visualization style of **pmlr-v151-liu22a**. Med-SVGd severely underestimates posterior covariance, whereas Ad-SVGd produces a substantially closer covariance approximation. Figure 6 further shows that, from an uninformed initialization of 1 in every component, Ad-SVGd learns component-wise bandwidths that differ from the initialization by several orders of magnitude.

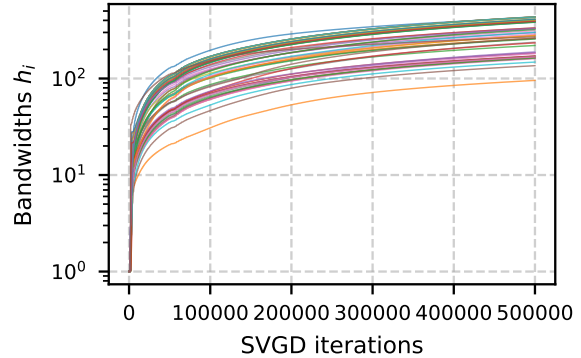


Figure 6: Bandwidth evolution in the Bayesian logistic-regression experiment.

6 Limitations and outlook

The principal analytical limitation is the reliance on Assumption 4, which abstracts the accuracy of the kernel-parameter optimization. A complete characterization of when the alternating gradient-ascent scheme in Algorithm 1 satisfies this assumption remains open. The theory therefore

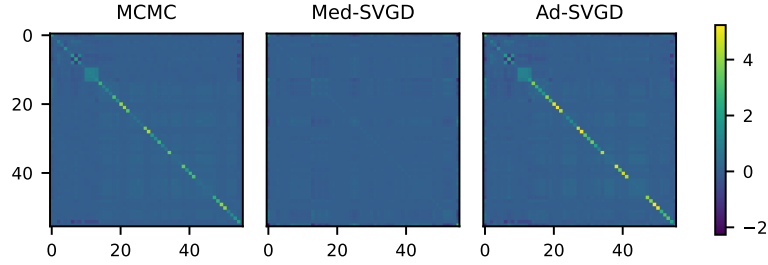


Figure 7: Bayesian logistic regression: covariance matrices of final SVGD particle sets compared with MCMC reference samples, averaged over 10 seeds.

establishes convergence under an explicit optimization-accuracy condition, while the experiments show that the practical gradient-ascent implementation is effective in the reported settings.

The theory is also mean-field and does not quantify finite-particle error, although the most visible empirical gains concern finite ensembles and variance recovery. In addition, the well-posedness of KSD maximization depends on the chosen parameter domain and regularity assumptions; the convergence statements should therefore be read together with the boundedness and uniform-kernel conditions stated in Section 4. The experiments focus primarily on comparisons with Med-SVGd, so they isolate the benefit of adaptive bandwidth selection but do not establish dominance over other adaptive or dimension-reduction variants of SVGd.

The analysis and experiments deliberately retain the original SVGd dynamics, which isolates kernel adaptation as the source of improvement. A natural next step is to combine the same KSD-driven kernel-selection mechanism with recent variants such as sliced SVGd [gong2021sliced], Grassmann SVGd [pmlr-v151-liu22a], or Stein transport [nsken2024steintransportbayesianinference], and to assess runtime-accuracy tradeoffs and sensitivity to the kernel-learning schedule more systematically.

A More numerical experiments

A.1 Gaussian mixture models

We revisit the example from Section 5.2. For the setup described there, Figure 9 shows histograms of the final particle distribution against the target density for $h \in \{0.001, 1, 1000\}$. We then compare Med-SVGD and Ad-SVGD across particle counts M , using final particle distributions after 10^4 iterations with step size 1.

Figure 8 reports the Wasserstein 1-distance between the final empirical particle distribution and the target distribution for Med-SVGD and

Figure 8: Wasserstein 1-approximation error for the one-dimensional Gaussian mixture.

Ad-SVGD with $M = 10, 20, 50, 100, 200, 500$. As expected, approximation quality improves with N for both methods. In this low-dimensional setting, the two approaches achieve comparable accuracy, reaching Wasserstein distance below 0.01 for $N = 500$.

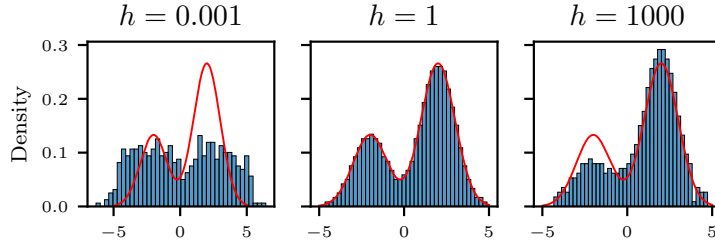
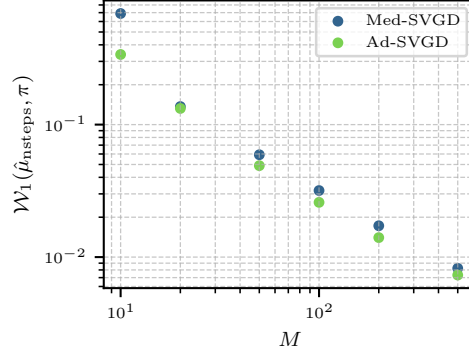


Figure 9: Histograms of final particle sets generated with fixed bandwidths $h = 0.001, 1, 1000$; the target density π is shown in red for comparison.

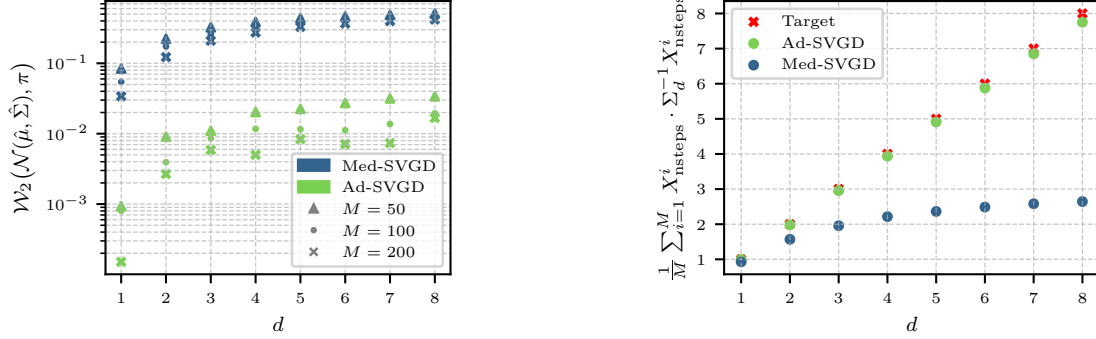
In the one-dimensional case, both methods approximate the target distribution well. The adaptive bandwidth strategy is stable, but the median heuristic is already sufficient in this simple setting, so Ad-SVGD provides no significant advantage here.

A.2 Scaling dimension: multivariate normal distribution

Moving to higher-dimensional controlled targets, we consider Gaussian distributions $\pi_d = \mathcal{N}(0, \Sigma_d)$ with $\Sigma_d = \text{diag}(1, \frac{1}{2}, \dots, \frac{1}{d})$ for $d \in \{2, \dots, 8\}$. The initial particle distribution is $\mathcal{N}(0, \frac{1}{d})^{\otimes d}$, and the algorithms are run for 10^4 iterations with step size 0.1. Whenever necessary for numerical stability, we use a smaller step size and adjust the number of iterations accordingly.

We compare the final particle approximations produced by Med-SVGD and Ad-SVGD as dimension increases. Because the Wasserstein 2-distance between Gaussian distributions has an explicit formula (see [StatisticalAspectsWasserstein]), sample quality is measured by the Wasserstein

2-distance between the target π_d and the Gaussian approximation $\mathcal{N}(\hat{\mu}, \hat{\Sigma})$, where $\hat{\mu}$ and $\hat{\Sigma}$ are the empirical mean and covariance of the particle set [StatisticalAspectsWasserstein]. Following the corresponding function in the Python Optimal Transport library [PythonOptimalTransport], used for these calculations, we refer to this quantity as the *Bures Wasserstein distance*. Figure 10a shows this error for Med-SVGD and Ad-SVGD with $M = 50, 100, 200$ as dimension increases. Across all reported dimensions, Ad-SVGD attains lower approximation error than Med-SVGD.



(a) Approximation error for $M \in \{50, 100, 200\}$.

(b) χ^2 -test statistic for $M = 200$.

Figure 10: Multivariate normal targets: approximation error and uncertainty diagnostic as dimension increases.

The Bures Wasserstein distance is relatively insensitive to the smaller variances in the later coordinates of π_d . To diagnose uncertainty recovery in those components, we also consider the statistic $X \cdot \Sigma_d^{-1} X$, which is χ^2 -distributed with d degrees of freedom, and hence has expectation d , when $X \sim \mathcal{N}(0, \Sigma_d)$ (cf. [Metropolisadjustedinteracting]). For $M = 200$, we compute the mean of this statistic over the final particle set and compare it with d in Figure 10b. As dimension increases, the Med-SVGD statistic moves progressively away from the target expectation, whereas the Ad-SVGD statistic remains comparatively close. This diagnostic identifies the same variance-underestimation behavior observed in the other experiments.

Tables 1 and 2 provide component-wise detail for the final particle distributions with $M = 200$. For each d , the tables list marginal particle variances alongside the target marginal variances shown in the first row. Table 1 reports Med-SVGD and Table 2 reports Ad-SVGD. Ad-SVGD closely matches the target marginal variances across components, whereas Med-SVGD substantially underestimates uncertainty, consistent with Figure 10b.

Finally, we examine marginal particle distributions for $M = 200$ and $d = 8$. For visualization, each component is normalized by the inverse of its target marginal standard deviation, equivalently multiplying the i -th component by i , so that each target marginal becomes standard normal. Figures 11 and 12 display the resulting normalized marginal histograms for Med-SVGD and Ad-SVGD, respectively, with a standard Gaussian density overlaid. Ad-SVGD captures the marginal distributions well, while Med-SVGD remains visibly underdispersed. The same conclusion appears in Figures 13 and 14, where normalized particle quantiles are compared with standard normal quantiles.

Table 1: Marginal variances of the final particle distribution for d -dimensional targets generated by Med-SVGD with $M = 200$; target marginal variances for π_d are shown for comparison.

component	1	2	3	4	5	6	7	8
Target	1.0000	0.2500	0.1111	0.0625	0.0400	0.0278	0.0204	0.0156
$d = 1$	0.9285							
$d = 2$	0.7921	0.1943						
$d = 3$	0.6803	0.1625	0.0697					
$d = 4$	0.6089	0.1440	0.0593	0.0311				
$d = 5$	0.5532	0.1275	0.0526	0.0271	0.0157			
$d = 6$	0.5190	0.1190	0.0481	0.0243	0.0140	0.0089		
$d = 7$	0.4900	0.1122	0.0449	0.0228	0.0131	0.0081	0.0052	
$d = 8$	0.4753	0.1077	0.0430	0.0215	0.0122	0.0074	0.0047	0.0032

Table 2: Marginal variances of the final particle distribution for d -dimensional targets generated by Ad-SVGD with $M = 200$; target marginal variances for π_d are shown for comparison.

component	1	2	3	4	5	6	7	8
Target	1.0000	0.2500	0.1111	0.0625	0.0400	0.0278	0.0204	0.0156
$d = 1$	0.9953							
$d = 2$	0.9907	0.2472						
$d = 3$	0.9867	0.2459	0.1095					
$d = 4$	0.9881	0.2467	0.1095	0.0610				
$d = 5$	0.9840	0.2433	0.1096	0.0616	0.0392			
$d = 6$	0.9858	0.2459	0.1090	0.0611	0.0392	0.0269		
$d = 7$	0.9856	0.2463	0.1086	0.0613	0.0390	0.0269	0.0199	
$d = 8$	0.9691	0.2409	0.1085	0.0611	0.0390	0.0268	0.0196	0.0150

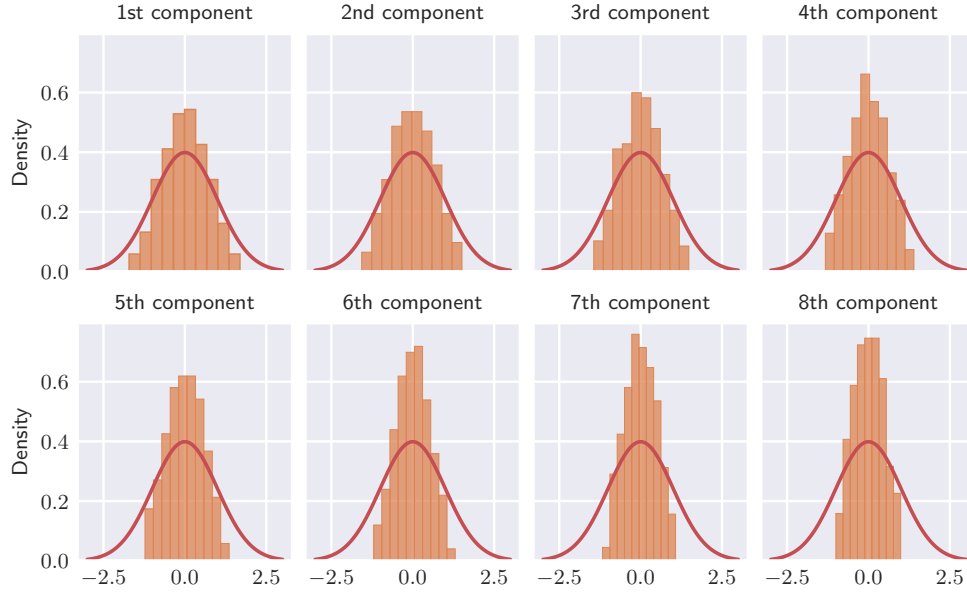


Figure 11: Normalized marginal histograms for the final Med-SVGD particle set in the eight-dimensional target with $M = 200$; the standard Gaussian density is shown for comparison.

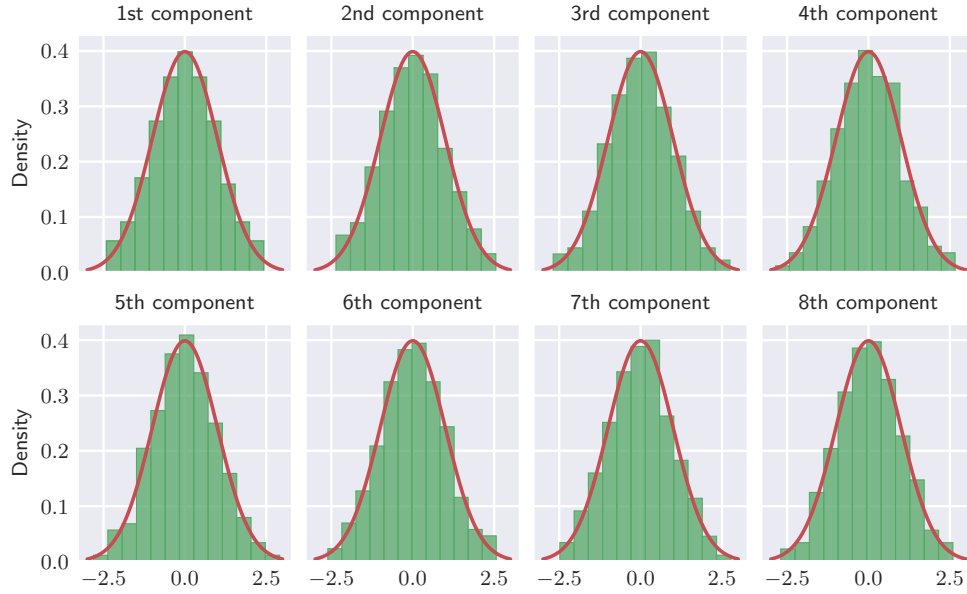


Figure 12: Normalized marginal histograms for the final Ad-SVGD particle set in the eight-dimensional target with $M = 200$; the standard Gaussian density is shown for comparison.

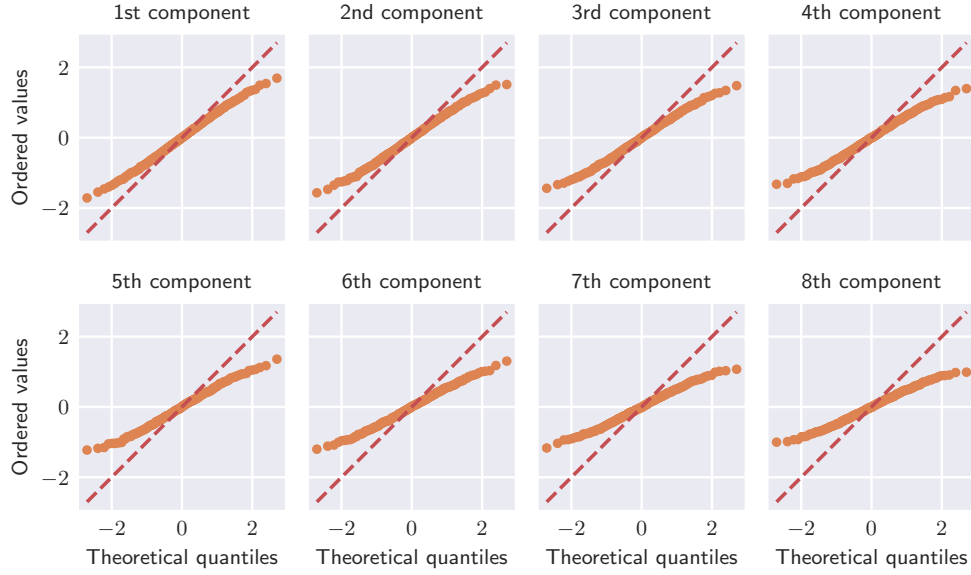


Figure 13: Q-Q plots comparing normalized marginals of the final Med-SVGd particle distribution for the eight-dimensional target with a standard normal distribution; the line of slope 1 through the origin is shown for comparison.

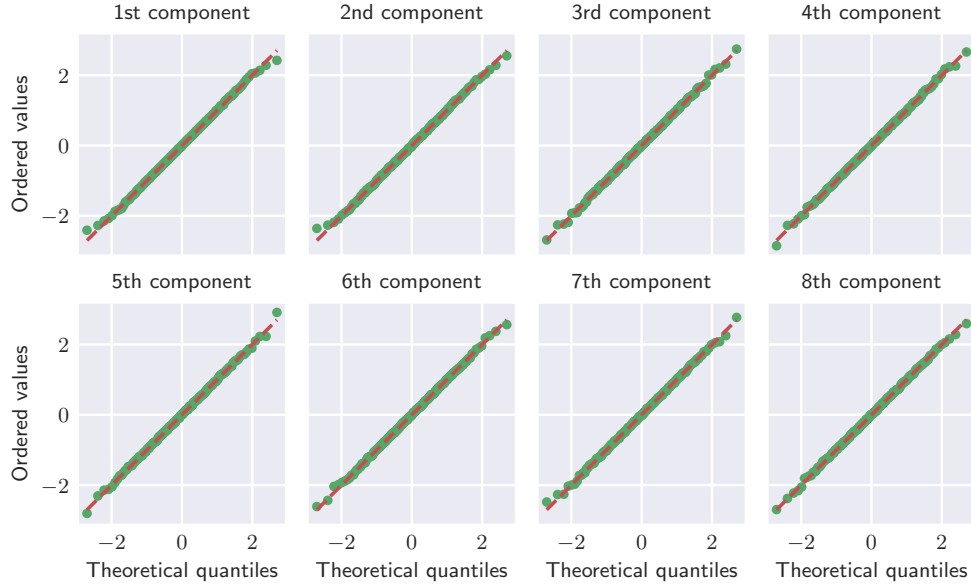


Figure 14: Q-Q plots comparing normalized marginals of the final Ad-SVGd particle distribution for the eight-dimensional target with a standard normal distribution; the line of slope 1 through the origin is shown for comparison.

A.3 Gaussian process inference

We further examine Ad-SVGD in a Gaussian-process (GP) inference task proposed in [reich21], using this setting to assess behavior as the coefficient dimension changes. We consider a GP on $[0, 1]$ represented by a truncated Karhunen-Loève (KL) expansion $u(s, x) = \sum_{k=1}^{N_x} x_k \psi_k(s)$ with basis functions $\psi_k(s) = \sqrt{2} \sin(k\pi s)$, where $x = (x_1, \dots, x_{N_x})^\top$ is a vector of independent Gaussian random variables $x_k \sim \mathcal{N}(0, k^{-2})$. We observe the process at N_y equispaced points in $[0, 1]$ and infer the coefficients x_k . For fixed N_x and N_y , this is an inverse problem with forward model $Y = AX + \varepsilon$, where the k -th column of $A \in \mathbb{R}^{N_y \times N_x}$ is $(\psi_k(s_1), \dots, \psi_k(s_{N_y}))^\top$, $s_i = \frac{i}{N_y}$ for $i = 1, \dots, N_y$. The prior is $X \sim \mathcal{N}(0, \Sigma)$, where Σ is diagonal with entries k^{-2} , $k = 1, \dots, N_x$, and the observational noise is independent Gaussian, $\varepsilon \sim \mathcal{N}(0, I_{N_y})$. We construct reference observations $\bar{y} \in \mathbb{R}^{N_y}$ by drawing $\bar{x} \sim \mathcal{N}(0, \Sigma)$ and setting $\bar{y} = A\bar{x}$.

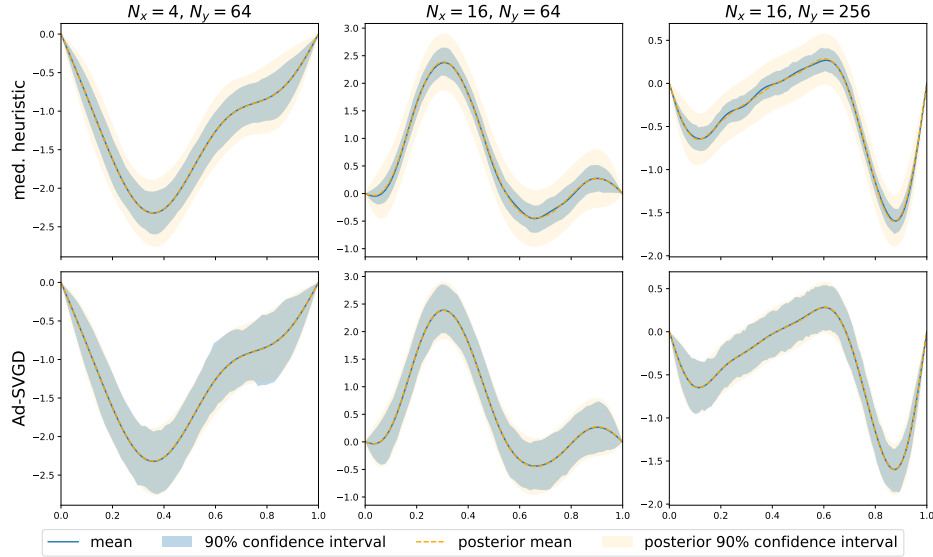


Figure 15: Estimated processes generated by median-heuristic SVGD and Ad-SVGD with $M = 100$ particles for different model configurations, compared with the posterior.

We use SVGD to approximate the resulting posterior $X|Y = \bar{y}$ and compare the median heuristic with the adaptive kernel strategy. For SVGD with the median heuristic, we use kernels of the form $k(x, y) = \exp(-\|x - y\|_1/h)$; for Ad-SVGD, we use product kernels $k(x, y) = \prod_{i=1}^{N_x} \exp(-|x_i - y_i|/h_i)$ with parameter $h = (h_1, \dots, h_{N_x})$, so each coefficient dimension has its own bandwidth. Following [SVGD], we use a variant of Adagrad for step-size control in the scenarios with $N_x = 16$.

Figure 15 compares the resulting process estimates with the posterior for $N_x = 4$ and $N_y = 64$, $N_x = 16$ and $N_y = 64$, and $N_x = 16$ and $N_y = 256$. As the coefficient dimension increases, median-heuristic SVGD underestimates posterior variance, whereas Ad-SVGD gives a closer uncertainty approximation. The same qualitative pattern appears for different numbers of process observations. Table 3 quantifies this behavior by comparing the covariance trace of the particle distributions with the exact posterior trace: the median heuristic severely underestimates uncertainty, while Ad-SVGD captures the variance more accurately.

N_x	4	8	16	16	16
N_y	64	64	64	128	256
theoretical	0.056	0.083	0.086	0.051	0.029
med. heuristic	0.026	0.023	0.022	0.012	0.006
Ad-SVGD	0.055	0.072	0.074	0.044	0.026

Table 3: Trace of the covariance of the final particle distribution ($M = 100$ particles) compared with the theoretical posterior trace for different configurations, averaged over 25 runs.

B Refined theoretical analysis

The KSD convergence result can be strengthened under a Stein log-Sobolev inequality, which relates KL divergence to KSD. **Duncan2019OnTG** introduced this inequality and discussed conditions under which it may hold. In the adaptive kernel-selection setting, the condition can be relaxed at the level of the family: it is sufficient that one kernel in the class provide the required log-Sobolev control. We formalize this requirement as follows.

Assumption 5. *Given a kernel class $\{k_\theta \mid \theta \in \Theta\}$, we assume that $\pi \in \mathcal{P}_1(\mathbb{R}^d)$ satisfies the generalized Stein log-Sobolev inequality with constant $\lambda > 0$:*

$$2\lambda \text{KL}(\mu \parallel \pi) \leq \max_{\theta \in \Theta} \text{KSD}_\theta^2(\mu \mid \pi)$$

for all $\mu \in \mathcal{P}_1(\mathbb{R}^d)$.

Under this property, convergence can be quantified directly in KL divergence. In particular, the following result gives an explicit bound on $\mathcal{F}(\mu_n)$ even when the optimization error is only uniformly bounded, $\varepsilon_n \leq \bar{\varepsilon}$, rather than converging to zero.

Theorem 4. *Suppose that Assumptions 1 to 3 and 5 are satisfied. For any $\alpha > 1$ and $\gamma > 0$ with*

$$\gamma \leq (\alpha - 1) \left(\alpha B(1 + \|\nabla V(0)\| + L \int_{\mathbb{R}^d} \|x\| d\pi(x) + L \sqrt{\frac{2\mathcal{F}(\mu_0)}{\lambda}} \right)^{-1},$$

and

$$\rho := 1 - 2\lambda c_\gamma \in (0, 1) \quad \text{where} \quad c_\gamma := \gamma \left(1 - \frac{\gamma B(\alpha^2 + L)}{2} \right),$$

it holds that

$$\mathcal{F}(\mu_{n+1}) \leq \rho \mathcal{F}(\mu_n) + c_\gamma \varepsilon_n,$$

where $(\mu_n)_{n \in \mathbb{N}}$ is generated by (10) and $\varepsilon_n \geq \max_{\theta \in \Theta} \text{KSD}_\theta^2(\mu_n \mid \pi) - \text{KSD}_{\theta_n}^2(\mu_n \mid \pi)$. In particular, if

1. $\varepsilon_n \leq \bar{\varepsilon}$ for all $n \in \mathbb{N}$, then

$$\mathcal{F}(\mu_n) \leq \rho^n \mathcal{F}(\mu_0) + \frac{c_\gamma \bar{\varepsilon}}{1 - \rho}$$

for all $n \in \mathbb{N}$.

2. $\varepsilon \leq c_\varepsilon q^n$ for some $q \in (0, 1)$, $c_\varepsilon > 0$ and all $n \in \mathbb{N}$. Under $\rho \neq q$ we have

$$\mathcal{F}(\mu_n) \in \mathcal{O}(\max\{\rho, q\}^n)$$

and under $\rho = q$ we have

$$\mathcal{F}(\mu_n) \in \mathcal{O}(\tilde{q}^n)$$

for all $\rho < \tilde{q} < 1$.

3. $\varepsilon_n \leq \frac{c_\varepsilon}{(n+1)^p}$ for $c_\varepsilon, p > 0$ and all $n \in \mathbb{N}$, then

$$\mathcal{F}(\mu_n) \leq \rho^n \mathcal{F}(\mu_0) + \frac{2^p c_\gamma c_\varepsilon}{n^p} \frac{1}{1 - \rho} + c_\gamma c_\varepsilon \frac{\rho^{\lfloor n/2 \rfloor}}{1 - \rho}$$

for all $n \in \mathbb{N}$.

Proof. As in the proof of Theorem 3, Assumption 5 and the stated conditions on α, γ give

$$\begin{aligned} \mathcal{F}(\mu_{n+1}) &\leq \mathcal{F}(\mu_n) - c_\gamma \max_{\theta \in \Theta} \text{KSD}_\theta^2(\mu_n \mid \pi) + c_\gamma \varepsilon_n \\ &\leq \rho \mathcal{F}(\mu_n) + c_\gamma \varepsilon_n, \end{aligned}$$

where $c_\gamma = \gamma(1 - \frac{\gamma B(\alpha^2 + L)}{2}) > 0$. Iterating this recursion over $n \in \mathbb{N}$ yields the general error bound

$$\mathcal{F}(\mu_n) \leq \rho^n \mathcal{F}(\mu_0) + c_\gamma \sum_{k=0}^{n-1} \rho^{n-1-k} \varepsilon_k.$$

We then treat the three stated regimes for ε_n separately:

1. Under $\varepsilon_n \leq \bar{\varepsilon}$, the first claim follows from the geometric-series bound

$$\sum_{k=0}^{n-1} \rho^{n-1-k} \leq \sum_{k=0}^{\infty} \rho^k = \frac{1}{1 - \rho}.$$

2. Assuming that $\varepsilon_n \leq c_\varepsilon q^n$ for some $q \in (0, 1)$ and $c_\varepsilon > 0$ we have

$$\mathcal{F}(\mu_n) \leq \rho^n \mathcal{F}(\mu_0) + c_\gamma c_\varepsilon \sum_{k=0}^{n-1} \rho^{n-1-k} q^k.$$

Suppose that $q > \rho$, then we have

$$\sum_{k=0}^{n-1} \rho^{n-1-k} q^k = \sum_{k=0}^{n-1} q^{n-1-k} \rho^k = q^{n-1} \sum_{k=0}^{n-1} (\rho/q)^k \leq q^{n-1} \frac{1}{1 - \rho/q} = q^n \frac{1}{q - \rho}$$

where we have used the formula for a geometric series. Similarly, in the case $q < \rho$, we directly bound

$$\sum_{k=0}^{n-1} \rho^{n-1-k} q^k = \rho^{n-1} \sum_{k=0}^{n-1} (q/\rho)^k \leq \rho^{n-1} \frac{1}{1 - q/\rho} = \rho^n \frac{1}{(\rho - q)}.$$

Finally, when $q = \rho$, we bound $q^k < \tilde{q}^k$ for any $q < \tilde{q} < 1$ and obtain the stated rate by the same argument.

3. In the setting of $\varepsilon_n \leq \frac{c_\varepsilon}{(n+1)^p}$, we rewrite the upper bound on $\mathcal{F}(\mu_n)$ as

$$\begin{aligned}
\mathcal{F}(\mu_n) &\leq \rho^n \mathcal{F}(\mu_0) + c_\gamma \sum_{k=0}^{n-1} \rho^k \varepsilon_{n-1-k} \\
&= \rho^n \mathcal{F}(\mu_0) + c_\gamma \left(\sum_{k=0}^{\lfloor n/2 \rfloor} \rho^k \varepsilon_{n-1-k} + \sum_{k=\lfloor n/2 \rfloor + 1}^{n-1} \rho^k \varepsilon_{n-1-k} \right) \\
&\leq \rho^n \mathcal{F}(\mu_0) + \frac{2^p c_\gamma c_\varepsilon}{n^p} \frac{1}{1-\rho} + c_\gamma c_\varepsilon \frac{\rho^{\lfloor n/2 \rfloor}}{1-\rho}.
\end{aligned}$$

Here we used $\varepsilon_{n-1-k} \leq \frac{c_\varepsilon}{(n-k)^p} \leq \frac{2^p c_\varepsilon}{n^p}$ for all $k \leq \lfloor n/2 \rfloor$ and $\varepsilon_k \leq c_\varepsilon$ for all $k \in \mathbb{N}$.

□